Day 22 Assignment

Description:
Implemented form handling, connected the backend to MongoDB, and added secure authentication with role-based access control.
This day focused on user registration, persistence, and protected routes.

Challenge 1 - Form Handling:
Created a registration form and processed submitted user data on the backend.
This demonstrated basic form submission handling in Express.

server.js

```
const express = require("express");
const bodyParser = require("body-parser");
const app = express();

app.use(bodyParser.urlencoded({ extended: true }));
app.use(bodyParser.json());

app.get("/register", (req, res) => {
  res.send(`
    <form method="POST" action="/register">
      <input type="text" name="name" placeholder="Enter name" required />
      <button type="submit">Register</button>
    </form>
  `);
});

app.post("/register", (req, res) => {
  const { name } = req.body;
  res.send(`Registration successful for ${name}`);
});

app.listen(4000, () => {
  console.log("Server running on http://localhost:4000");
});
```

localhost:4000/register

Hemanth Saladi | Register

localhost:4000/register

Registration successful for Hemanth Saladi

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Wipro\Assignments\Day 22 (Forms,Database,Auth)\Day22> node .\server.js
Server running on http://localhost:4000
```

Challenge 2 - Database Integration:
Connected the application to MongoDB using Mongoose and stored registered user details.
This enabled persistent storage of user data.

```
PS D:\Wipro\Assignments\Day 22 (Forms,Database,Auth)\Day22> node .\server.js
[dotenv@17.3.1] injecting env (2) from .env -- tip:
Server running on http://localhost:4000
MongoDB connected
User saved successfully: {
  name: 'Hemanth',
  email: 'Hemanth@gmail.com',
  password: 'user123',
  role: 'user',
  _id: new ObjectId('69bace304496de15d5ce307e'),
  __v: 0
}
```

localhost:4000/register

Hemanth | Hemanth@gmail.com | ••••

Register

localhost:4000/register

Registration successful for Hemanth

Saladi V S S Siva Hemanth Kumar
hemanthsaladi2004@gmail.com
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Challenge 3 - Authentication and RBAC:
Implemented Passport.js authentication with sessions and restricted admin route access based on user role. This added secure login and role-based protection.

```
app.post("/login",
  passport.authenticate("local", {
    failureRedirect: "/login-failure"
  }),
  (req, res) => {
    res.send("Login successful");
  }
);

app.get("/login-failure", (req, res) => {
  res.send("Login failed");
});

function isAdmin(req, res, next) {
  if (req.isAuthenticated() && req.user.role === "admin") {
    return next();
  }
  return res.status(403).send("Access Denied");
}

app.get("/admin", isAdmin, (req, res) => {
  res.send("Welcome, Admin!");
});

app.listen(4000, () => {
  console.log("Server running on http://localhost:4000");
});
```

Note:
MongoDB must be running locally, or you must use a MongoDB Atlas connection string in .env

Saladi V S S Siva Hemanth Kumar
hemanthsaladi2004@gmail.com
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